CYBERSRS

Secure Campus Wi-Fi Management System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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Contents

1. Introduction 3

2. Functional requirements 4

3. Non-functional requirements 6

4. Security, privacy and compliance 7

5. Data requirements 7

6. Network and interfaces 7

7. Compliance and policy applicability matrix 7

8. Architecture 7

9. Threats 11

10. Risks 12

11. User stories and use cases 12

12. Verification and acceptance 12

13. Glossary 14

14. SDLC recommendation and workflow 15

SDLC rankings and decision evidence 18

Evidence register 19

Source references 19

Validation findings 20

Human review history 21

AI drafting record 21

Requirements traceability matrix 22

1. Introduction

Original project description

Design a secure campus Wi-Fi management system for a university that provides wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices. The system should use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts. Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs. The solution should support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure campus Wi-Fi management system for a university that provides wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The solution should support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.

Status: confirmed

Evidence ids: E-004

Open issues: How will device-level policies be enforced on personal devices?; What specific thresholds or triggers define repeated authentication failures or unauthorized access attempts?; How

will audit logs be stored, accessed, and retained for compliance purposes?

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.

Statement: The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A user attempts to connect to the campus Wi-Fi in an academic building

When: The system processes the connection request

Then: The system establishes a secure wireless session with valid authentication credentials

Method: test

Observable evidence: Test capture of successful connection establishment

Unresolved decisions: Definition of 'secure' in context of campus Wi-Fi

FR-002 - use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Statement: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A user with invalid credentials repeatedly attempts to authenticate

When: The system processes the authentication sequence

Then: The system detects repeated failure and generates a security alert

Method: test

Observable evidence: Test capture of generated security alert

Given: A device connects to the network without proper authorization

When: The system monitors active sessions

Then: The system identifies unauthorized access and triggers an alert

Method: test

Observable evidence: Test capture of session detection and alert generation

Unresolved decisions: Threshold for defining 'repeated' authentication failures; Definition of ' unauthorized access attempts'

FR-003 - Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Statement: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrator accesses the management interface

When: The administrator configures a new wireless policy

Then: The system applies the policy to active network segments

Method: test

Observable evidence: Test capture of policy configuration and applied state

Given: An administrator requests device list from a network segment

When: The system retrieves connected devices

Then: The system returns a list of active devices with associated identifiers

Method: inspection

Observable evidence: Test capture of device list response

Given: An administrator initiates access revocation for a device

When: The system processes the revoke request

Then: The system processes the revoke request

Method: demonstration

Observable evidence: Test capture of permission state

Given: An administrator requests audit logs for a time period

When: The system retrieves the log entries

Then: The system returns the requested audit records in chronological order

Method: inspection

Observable evidence: Test capture of retrieved audit log content

Unresolved decisions: Scope and format of accessible audit logs; Definition of 'connected devices' for policy review

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - The solution shall support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.

Statement: The solution shall support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A university user with a personally owned device

When: The device connects to the wireless network using university-approved configuration

Then: The connection is authenticated and the device is registered in the system's device inventory

Method: test

Observable evidence: Test capture of device registration response

Given: A campus location with multiple buildings and high device density

When: The network handles concurrent connections from personal devices under load

Then: No performance degradation or connectivity loss occurs during peak usage

Method: demonstration

Observable evidence: Performance test log showing stable throughput and latency

Unresolved decisions: How device approval is validated; Whether personal device data is collected or shared with university systems

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-002; FR-003

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-003; NFR-001

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003

7. Compliance and policy applicability matrix

Open decisions: How will device-level policies be enforced on personal devices?; What specific thresholds or triggers define repeated authentication failures or unauthorized access attempts?; How will audit logs be stored, accessed, and retained for compliance purposes?

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Wireless Access Policy Engine

Detail: Proposed logical owner for FR-002, NFR-001, FR-001.

Status: proposed

Evidence ids: E-001; E-002; E-004

Requirement ids: FR-002; NFR-001; FR-001

Component name: Wireless Access Policy Engine

Responsibility: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts. The solution shall support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations. The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and

administrative offices.

Authentication and Authorization Service

Detail: Proposed logical owner for FR-002, FR-003, FR-001.

Status: proposed

Evidence ids: E-001; E-002; E-003

Requirement ids: FR-002; FR-003; FR-001

Component name: Authentication and Authorization Service

Responsibility: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts. Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs. The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.

Session Monitoring and Threat Detection Module

Detail: Proposed logical owner for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Component name: Session Monitoring and Threat Detection Module

Responsibility: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Network Administrator Console

Detail: Proposed logical owner for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Component name: Network Administrator Console

Responsibility: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-002.

Status: proposed

Evidence ids: E-001; E-002

Requirement ids: FR-001; FR-002

From component: Wireless Access Policy Engine

To component: Authentication and Authorization Service

Data or action: The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices. The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

From component: Wireless Access Policy Engine

To component: Session Monitoring and Threat Detection Module

Data or action: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-002.

Status: proposed

Evidence ids: E-001; E-002

Requirement ids: FR-001; FR-002

From component: Authentication and Authorization Service

To component: Wireless Access Policy Engine

Data or action: The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices. The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

From component: Authentication and Authorization Service

To component: Session Monitoring and Threat Detection Module

Data or action: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined interaction

Detail: Proposed directed path for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

From component: Authentication and Authorization Service

To component: Network Administrator Console

Data or action: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

From component: Session Monitoring and Threat Detection Module

To component: Wireless Access Policy Engine

Data or action: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

From component: Session Monitoring and Threat Detection Module

To component: Authentication and Authorization Service

Data or action: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined interaction

Detail: Proposed directed path for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

From component: Network Administrator Console

To component: Authentication and Authorization Service

Data or action: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-001, FR-002.

Status: proposed

Evidence ids: E-001; E-002

Requirement ids: FR-001; FR-002

Untrusted component: Wireless Access Policy Engine

Trusted component: Authentication and Authorization Service

Enforcement: The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices. The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Untrusted component: Session Monitoring and Threat Detection Module

Trusted component: Authentication and Authorization Service

Enforcement: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A user attempts to connect to the campus Wi-Fi in an academic building
2. When: The system processes the connection request

Expected result: Expected: The system establishes a secure wireless session with valid authentication credentials

Evidence to record: Evidence: Test capture of successful connection establishment

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: A user with invalid credentials repeatedly attempts to authenticate
2. When: The system processes the authentication sequence

Expected result: Expected: The system detects repeated failure and generates a security alert

Evidence to record: Evidence: Test capture of generated security alert

Test type: positive

Id: TEST-003

Requirement ids: FR-002

Procedure

1. Given: A device connects to the network without proper authorization
2. When: The system monitors active sessions

Expected result: Expected: The system identifies unauthorized access and triggers an alert

Evidence to record: Evidence: Test capture of session detection and alert generation

Test type: negative

Id: TEST-004

Requirement ids: FR-003

Procedure

1. Given: An administrator accesses the management interface
2. When: The administrator configures a new wireless policy

Expected result: Expected: The system applies the policy to active network segments

Evidence to record: Evidence: Test capture of policy configuration and applied state

Test type: positive

Id: TEST-005

Requirement ids: FR-003

Procedure

1. Given: An administrator requests device list from a network segment
2. When: The system retrieves connected devices

Expected result: Expected: The system returns a list of active devices with associated identifiers

Evidence to record: Evidence: Test capture of device list response

Test type: inspection

Id: TEST-006

Requirement ids: FR-003

Procedure

1. Given: An administrator initiates access revocation for a device
2. When: The system processes the revoke request

Expected result: Expected: The system processes the revoke request

Evidence to record: Evidence: Test capture of permission state

Test type: positive

Id: TEST-007

Requirement ids: FR-003

Procedure

1. Given: An administrator requests audit logs for a time period
2. When: The system retrieves the log entries

Expected result: Expected: The system returns the requested audit records in chronological order

Evidence to record: Evidence: Test capture of retrieved audit log content

Test type: inspection

Id: TEST-008

Requirement ids: NFR-001

Procedure

1. Given: A university user with a personally owned device
2. When: The device connects to the wireless network using university-approved configuration

Expected result: Expected: The connection is authenticated and the device is registered in the system's device inventory

Evidence to record: Evidence: Test capture of device registration response

Test type: positive

Id: TEST-009

Requirement ids: NFR-001

Procedure

1. Given: A campus location with multiple buildings and high device density
2. When: The network handles concurrent connections from personal devices under load

Expected result: Expected: No performance degradation or connectivity loss occurs during peak usage

Evidence to record: Evidence: Performance test log showing stable throughput and latency

Test type: negative

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Access control

Detail: Rules and enforcement governing which principals may perform actions on resources.

Status: proposed

Evidence ids: E-002

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

14. SDLC recommendation and workflow

Selected model: Agile

Justification: The selection of Agile as the required model is based on a uniform score of 40 across all candidates, with no differentiating input evidence provided to favor one methodology over another. Given the absence of explicit factors such as budget, delivery timelines, failure consequences, feedback mechanisms, legacy systems, regulation, schedule constraints, stability needs, uncertainty levels, or verification requirements, no candidate model can be objectively ranked for suitability. The equal scores indicate a lack of evidence to support any specific lifecycle model. As a result, Agile is provisionally selected due to its inherent adaptability and iterative nature—particularly relevant in dynamic environments like university campuses where evolving access needs and security threats require responsive adjustments. However, this decision remains provisional pending resolution of the unknown factors listed. The existing requirements emphasize secure wireless access, centralized authentication, role-based controls, monitoring, alerts, policy configuration, device inspection, and scalability—all of which align with Agile's iterative delivery and continuous feedback loops. Without evidence to differentiate among models, no model is objectively superior; thus, the selection is not justified by data but by default alignment with operational flexibility. The previous validation failure—work_groups_must_not_duplicate_requirements—indicates a need for explicit requirement grouping to avoid redundancy, which must be addressed in implementation.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.; FR-002: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.; FR-003: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.
2. Review and baseline the source-defined work group: NFR-001: The solution shall support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline

2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall provide secure wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.; FR-002: The system shall use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.; FR-003: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.
3. Implement the source-defined work group: NFR-001: The solution shall support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.
4. In an authorized isolated test environment, execute TEST-001: Given: A user attempts to connect to the campus Wi-Fi in an academic building When: The system processes the connection request Expected: The system establishes a secure wireless session with valid authentication credentials
5. In an authorized isolated test environment, execute TEST-002: Given: A user with invalid credentials repeatedly attempts to authenticate When: The system processes the authentication sequence Expected: The system detects repeated failure and generates a security alert
6. In an authorized isolated test environment, execute TEST-003: Given: A device connects to the network without proper authorization When: The system monitors active sessions Expected: The system identifies unauthorized access and triggers an alert
7. In an authorized isolated test environment, execute TEST-004: Given: An administrator accesses the management interface When: The administrator configures a new wireless policy Expected: The system applies the policy to active network segments
8. In an authorized isolated test environment, execute TEST-005: Given: An administrator requests device list from a network segment When: The system retrieves connected devices Expected: The system returns a list of active devices with associated identifiers
9. In an authorized isolated test environment, execute TEST-006: Given: An administrator initiates access revocation for a device When: The system processes the revoke request Expected: The system processes the revoke request

10. In an authorized isolated test environment, execute TEST-007: Given: An administrator requests audit logs for a time period When: The system retrieves the log entries Expected: The system returns the requested audit records in chronological order
11. In an authorized isolated test environment, execute TEST-008: Given: A university user with a personally owned device When: The device connects to the wireless network using university-approved configuration Expected: The connection is authenticated and the device is registered in the system's device inventory
12. In an authorized isolated test environment, execute TEST-009: Given: A campus location with multiple buildings and high device density When: The network handles concurrent connections from personal devices under load Expected: No performance degradation or connectivity loss occurs during peak usage
13. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
14. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure campus Wi-Fi management system for a university that provides wireless access to students, faculty, staff, and guests across academic buildings, hostels, laboratories, and administrative offices.

Id: E-002

Source: description

Quote: The system should use centralized authentication, role-based access control, separate guest access from internal university resources, monitor active wireless sessions, detect repeated authentication failures and unauthorized access attempts, and generate security alerts.

Id: E-003

Source: description

Quote: Network administrators should be able to configure wireless access policies, review connected devices, revoke access, and inspect audit logs.

Id: E-004

Source: description

Quote: The solution should support university-managed and approved personal devices and remain secure, scalable, reliable, and manageable across multiple campus locations.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.6857

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.6818

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:7 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 7

Page or section: text chunk 8

Relevance score: 0.6703

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.6767

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-92ffa375cd05a8f5:16 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.6789

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	TEST-004, TEST-005, TEST-006, TEST-007, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-004	TEST-008, TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28